SDC

Hackathon

Phase 1: Preliminary

1.0 TEAM AND PROJECT OVERVIEW

Team Name: kpmbians

Selected Domain: Domain 2 – Orbyte Smart SME Challenge

Selected SDG Track (1–17): SDG 8 – Decent Work and Economic Growth

Project / Solution Name: TwinBoss

Name (Leader): NUR ALIA MAISARAH BINTI MUHAMMAD AMIN

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University Affiliation(s): Kolej Profesional MARA Beranang

Number of Team Members: 2

Team Members:

1. Nur Alia Maisarah binti Muhammad Amin (Team Leader)
2. Nur Husnina Hidayah binti Abdul Razi

2.0 EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1 The Problem Context

Micro and small enterprise (SME) owners often face operational challenges when managing customer refund requests, particularly when handling multiple orders, verifying complaints, and making financial decisions. In businesses with limited staff and resources, refund management can become a time-consuming process that requires owners to personally review requests, communicate with customers, and authorize transactions.

The problem becomes more complicated when refund claims involve damaged products, missing items, or potentially fraudulent requests. Without a structured verification process, business owners may struggle to distinguish legitimate complaints from suspicious claims. Inconsistent documentation and reliance on manual communication channels can also make it difficult to track decisions, maintain records, and prevent duplicate refund requests.

These challenges can create unnecessary administrative workloads, delay customer service, and expose SMEs to avoidable financial losses. Owners may also find it difficult to balance customer satisfaction with cash-flow protection, especially when deciding whether to issue cash refunds or offer alternative compensation.

Although artificial intelligence can assist with information analysis and routine decision-making, SMEs need solutions that preserve human oversight over sensitive financial decisions. An effective system should therefore help business owners evaluate refund requests more consistently, identify potential risks, and delegate routine tasks without completely surrendering control.

TwinBoss addresses this operational challenge by exploring how AI-assisted refund assessment and structured approval workflows can help SMEs reduce administrative burdens, improve refund management, and make more informed business decisions.

2.2 The Proposed AI Solution

TwinBoss is an AI-powered refund management solution designed to help small and medium-sized enterprises streamline refund assessment and reduce repetitive administrative work. The system introduces a structured workflow involving three main roles: customers, staff, and business owners.

Customers submit refund requests with their order information, a description of the issue, and supporting photographic evidence where applicable. Staff then review the requests with assistance from an AI assessment engine that evaluates the submitted information and identifies potential inconsistencies based on the available business records.

The system also incorporates a configurable set of business rules known as the Business Constitution. These rules help guide refund recommendations based on factors such as customer status, refund value, and business cash-flow considerations. This allows the system to support more consistent decisions while maintaining appropriate human oversight.

The workflow separates assessment, verification, approval, and execution. Staff review the request, the business owner authorizes the decision, and authorized staff proceed with the subsequent refund workflow and customer communication.

TwinBoss is developed using React, TypeScript, and Tailwind CSS, with a large language model accessed through the Groq API to support AI-assisted assessment. Synthetic mock business data simulates the Orbyte business environment for demonstration purposes; the prototype does not assume integration with a live Orbyte database.

The solution aims to reduce administrative burdens, improve accountability, and help SMEs balance customer satisfaction with financial control. AI supports the decision-making process rather than replacing human judgement.

3.0 TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1 Conceptual Tech

Technologies, Frameworks, APIs, and Databases

TwinBoss is designed as a web-based application that integrates an AI-assisted decision-making engine with a structured refund management workflow. The proposed technology stack consists of the following components:

Component	Technology	Purpose
Frontend	React 18	Provides the user interface for customers, staff, and business owners.
Build Tool	Vite	Supports frontend development and application bundling.
Programming Language	TypeScript	Implements application logic and provides static type checking.
Styling	Tailwind CSS	Supports responsive layouts and consistent interface styling.
AI Engine	Groq API with an LLM	Assists with refund assessment, risk identification, and business-rule-based recommendations.
Model	openai/gpt-oss—20b via Groq	Intended to generate structured AI

		assessments and recommendations.
State Management	React useState	Manages application state during the demonstration.
Business Data Layer	Local mock data (mockData.ts)	Simulates order records and customer information for the prototype.
Icon Library	Lucide Icons	Provides interface icons and visual indicators.
Development and Hosting	Replit and GitHub	Supports development, code management, and potential deployment, subject to the final setup.

Conceptual System Architecture

TwinBoss follows a layered architecture in which the user interface interacts with application logic, an AI assessment service, and a simulated business-data layer.

The customer submits a refund request through the frontend. The application retrieves the relevant simulated order information and supplies the necessary context to the AI engine. The AI generates an assessment that helps staff evaluate the request and provides a recommendation for the business owner. Following authorization, staff can proceed with the next steps in the refund workflow.

The prototype uses simulated business records to demonstrate the proposed integration with the Orbyte ecosystem. Actual Orbyte API integration is not assumed in this proposal.

System Flow:

Customer Portal → Refund Request Processing → Mock Business Data and AI Assessment → Staff Verification → Business Owner Approval → Staff Execution and Customer Notification

SDC Sponsor-Provided Resources

The project is intended to demonstrate AI-assisted business operations using an LLM API and a web-based development environment. The Groq API is identified in the current project plan as the AI inference service, while Replit and GitHub may support development, code management, or deployment.

The team will identify any officially sponsor-provided API or cloud platform actually used during development and document its specific role. The use of a third-party API or development platform alone does not establish that it was provided or sponsored by SDC.

3.2 Design-Thinking & Deployment Strategy

TwinBoss applies a human-centred design approach to address the operational challenges faced by SME owners when managing refund requests. The solution focuses on reducing repetitive administrative tasks while maintaining appropriate human oversight over financial decisions.

The proposed workflow separates refund assessment, verification, authorization, and execution into distinct stages involving customers, staff, and business owners. This Maker-Checker approach is intended to improve accountability and reduce the risk of unauthorized refund decisions. AI-assisted assessment and configurable business rules will support consistent recommendations, while the final approval process preserves the owner's control over business finances.

For usability, TwinBoss is designed as a web-based application with role-specific interfaces and a structured workflow. This approach aims to make refund management easier to understand and operate without requiring users to navigate fragmented communication channels.

During Phase 1, the team intends to validate the core workflow using simulated business data and an AI inference API. For Phase 2, development priorities include testing the reliability of AI-generated assessments, strengthening API-key security through backend integration where necessary, improving error handling, and evaluating the system with representative refund scenarios. Further testing will also assess usability, processing time, and the consistency of recommendations before considering deployment in a real SME environment.

4.0 TECHNICAL PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

1. **Target SDG Sub-Target(s)**

SDG 8: Decent Work and Economic Growth — Target 8.3

TwinBoss aligns with Target 8.3, which promotes productive activities, entrepreneurship, creativity, innovation, and the growth of micro, small, and medium-sized enterprises. By streamlining refund management and supporting more consistent operational decision-making, TwinBoss aims to help SMEs reduce administrative burdens and allocate more time to core business activities.

2. **Current Problem Baseline**

Preliminary estimate: 10–15 minutes per refund request.

For initial planning purposes, TwinBoss uses an estimated baseline of 10–15 minutes for manually reviewing a refund request, checking order information, communicating with relevant parties, and obtaining approval. This figure is a working assumption rather than a measured industry average and will be validated through representative workflow testing during Phase 2.

3. **Core Measurable KPI**

Percentage reduction in average refund-processing time per request.

Processing time will be measured from the start of refund-request review until an approval decision is reached. The team will compare the average time required using the manual workflow with the time required using TwinBoss under comparable test conditions.

4. Phase 2 Impact Target

Target: Reduce average refund-assessment and approval time by 30% compared with the validated manual baseline.

This preliminary target will be evaluated through controlled test scenarios using representative refund requests. The results will help determine whether AI-assisted assessment and a structured approval workflow can reduce administrative effort while maintaining appropriate human oversight and decision quality.

5.0 ELIGIBILITY & READINESS CHECKLIST

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6.0 IP & ORIGIANLITY DECLARATION

6.1 Pre-Existing / Prior-Ownes Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

Answer: The TwinBoss project is developed for the SDC Hackathon 2026. No pre-existing code, designs, research, or other previously owned work are included in this submission.

6.2 Third-Party Code, Datasets or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

Answer: Yes. TwinBoss uses third-party open-source frameworks, libraries, and a pre-trained AI model to support its development and AI-assisted functionality. The project uses the following resources:

Name	Type (Code/Dataset/Model)	License
React 18	Frontend Library	MIT
Vite	Build Tool	MIT
TypeScript	Programming language and tooling	Apache 2.0
Tailwind CSS	CSS Framework	MIT
Lucide Icons	Icon library	ISC
Openai/gpt-oos-20b via Groq API	Pre-trained AI model	Apache 2.0

Datasets: No public or private real-world datasets are used in the project. All mock business data used for demonstration and testing are synthetically generated by the team. No real customer, financial, or business records are included.

The team acknowledges the use of third-party resources and will comply with their applicable licensing and usage requirements.

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

Answer: No. To the team's knowledge, neither team member is subject to an employment, internship, or consulting agreement that would grant an employer ownership rights over the intellectual property developed for the SDC Hackathon 2026.

7.0 DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC Hackathon 2026 period.

Team Leader: NUR ALIA MAISARAH BINTI MUHAMMAD AMIN

Project Name: TwinBoss

Team Name: kpmbians

Signature: 

Date: 24/10/2026